From variational Monte Carlo to superfluid hydrodynamics in microwave-shielded Bose gases

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We show that time-dependent variational Monte Carlo (tVMC) based on a Jastrow–Feenberg wavefunction, when the pair correlations are frozen at their ground-state values, reduces *exactly* to irrotational superfluid hydrodynamics with a microscopically determined equation of state (EOS). The EOS is extracted from a single variational Monte Carlo calculation on the uniform system and encodes the full effect of two-body correlations into a density-dependent chemical potential $\mu(\rho)$. Recasting the resulting hydrodynamic equations as a generalized Gross–Pitaevskii equation (GPE), we reproduce the breathing-mode dynamics of a trapped microwave-shielded dipolar Bose gas—obtained from the full many-body tVMC—to within 1% across particle numbers $N = 8$ to $N = 128$, at a computational cost reduced by orders of magnitude. The framework provides a practical, quantitatively reliable tool for predicting collective modes in the rapidly growing class of microwave-shielded molecular experiments.

Introduction.—Microwave-shielded polar molecules have emerged as a versatile platform for quantum many-body physics, with recent experiments achieving Bose–Einstein condensation [1], observing shielded collisions [2, 3], and exploring novel dipolar phases [4–6]. A central challenge is to predict the collective dynamics of these systems—breathing modes, expansion dynamics, quench responses—starting from the microscopic interaction potential, which combines long-range dipolar attraction ($\sim C_3/r^3$) with short-range repulsive shielding ($\sim C_6/r^6$).

Mean-field Gross–Pitaevskii (GP) theory, while computationally efficient, relies on a contact-interaction approximation that poorly captures the anisotropic, density-dependent correlations of shielded molecules. On the other hand, *ab initio* many-body methods such as time-dependent variational Monte Carlo (tVMC) [7, 8] can treat realistic interactions and correlations but require stochastic sampling at every time step, severely limiting accessible timescales and system sizes.

In this Letter, we bridge this gap by establishing an *exact* connection between correlated tVMC dynamics and deterministic superfluid hydrodynamics. We show that freezing the two-body Jastrow correlations reduces the many-body time-dependent variational principle (TDVP) to continuity and Euler equations whose equation of state (EOS) is microscopically determined by a one-time VMC calculation on the uniform system. The resulting generalized GPE—with the VMC-derived EOS replacing the standard contact coupling $g|\psi|^2$—reproduces the full tVMC breathing dynamics to $< 1\%$ accuracy for $N = 8$ and $N = 128$ microwave-shielded bosons, while reducing computational cost by two orders of magnitude.

Model.—We consider N bosons in a cylindrically symmetric harmonic trap with Hamiltonian

$$\hat{H}(t) = \sum_{i=1}^{N}\big[-\tfrac{1}{2}\nabla_i^2 + V_{\rm tr}(\mathbf{r}_i, t)\big] + \sum_{i<j} v(\mathbf{r}_{ij}), \qquad (1)$$

where $V_{\rm tr} = \frac{1}{2}[\omega_\rho(t)^2\rho^2 + \omega_z^2 z^2]$ (natural units $\hbar = m = 1$) and the two-body interaction models microwave-shielded dipolar molecules:

$$v(\mathbf{r}) = \frac{C_3}{r^3}(3\cos^2\theta - 1) + \frac{C_6}{r^6}(\sin^4\theta + w_{12}\sin^2\theta\cos^2\theta). \quad (2)$$

This effective potential, derived from Floquet scattering theory for dual-microwave-dressed molecules [9], is parametrized by coefficients C_3, C_6, and w_{12} that are determined by the microwave field configuration. In a realistic experiment these are fixed by the dressing scheme; here, for demonstration, we choose $C_3 = 10^{-3}$, $C_6 = 10^{-6}$, and $w_{12} = 2.0$ in oscillator units. We study a quench $\omega_\rho : 1 \to 2$ at $t = 0$ that excites radial breathing modes.

tVMC with Jastrow–Feenberg ansatz.—The many-body wavefunction is parametrized as a truncated Jastrow–Feenberg [10, 11] ansatz:

$$\Psi(\mathbf{X}, t) = \exp\Big[\sum_i u_1(\mathbf{r}_i, t) + \sum_{i<j} u_2(r_{ij}, \hat{\mathbf{r}}_{ij})\Big], \qquad (3)$$

where $u_1(\mathbf{r}, t)$ is a complex, time-dependent one-body field and u_2 encodes two-body correlations.

One-body field.—In cylindrical coordinates, u_1 is parametrized with a separable structure $u_1 = u_\rho(\rho, t) + u_z(z, t)$, where each component consists of a harmonic reference plus Gaussian corrections with complex, time-dependent coefficients [12].

Two-body Jastrow.—The pair factor is expanded in partial waves. The interaction (2) couples angular momenta up to $\ell = 4$ (through the $\sin^4\theta$ term in the shielding potential [9]):

$$u_2(r, \cos\theta) = u_{\rm cusp}(r) + \sum_{\ell=0,2,4} f_\ell(r)\, P_\ell(\cos\theta), \qquad (4)$$

where $u_{\rm cusp}(r)$ captures the short-range scattering boundary condition and each f_ℓ is a Gaussian expansion [12]. For the interaction parameters used here, the $\ell = 4$ component is negligible ($|c_4|_{\rm rms}/|c_0|_{\rm rms} = 0.07\%$) and even the $\ell = 2$ anisotropy is minor (3% of $\ell = 0$); we therefore retain only $\ell = 0$ and $\ell = 2$ in practice [12].

Figure 1 shows the quality of the optimized pair correlations. The leading-order hypernetted-chain (HNC) approximation to the pair distribution function, $g_2^{\rm HNC}(r, \theta) = e^{2u_2(r,\cos\theta)}$ [13], reveals a correlation hole at short range [Fig. 1(a)], reflecting the interplay between dipolar attraction along $\theta = 0$ and the shielding repulsion. The pair correlation is nearly isotropic, consistent with the subdominant role of f_2.

Equations of motion.—The TDVP [14, 15] projects the Schrödinger equation onto the variational tangent space. Defining logarithmic derivatives $O_\alpha = \partial_{\theta_\alpha} \ln \Psi$ and the local energy $E_L = (\hat{H}\Psi)/\Psi$, the equations of motion read [7]

$$\sum_\beta S_{\alpha\beta}\, \dot{\theta}_\beta = -i\, F_\alpha, \qquad (5)$$

where the overlap matrix and force vector are

$$S_{\alpha\beta} = \langle \Delta O_\alpha^* \, \Delta O_\beta \rangle_t, \qquad (6)$$
$$F_\alpha = \langle \Delta O_\alpha^* \, \Delta E_L \rangle_t, \qquad (7)$$

with $\Delta A = A - \langle A \rangle_t$ denoting connected fluctuations. The expectation values are estimated stochastically by sampling from $|\Psi|^2$ using Metropolis Monte Carlo with $N_w = 65{,}536$ GPU-accelerated walkers [16]. The linear system (5) is solved via eigendecomposition of S with truncation at $\lambda_{\rm min}/\lambda_{\rm max} = 10^{-4}$ and a diagonal shift $\eta = 10^{-2}$ for regularization. Time integration uses a fourth-order Runge–Kutta (RK4) scheme, with each stage requiring independent MC sampling. Full details of the MC procedure and convergence diagnostics are given in Ref. [12].

Frozen-u_2 ⇒ superfluid hydrodynamics.—Our central theoretical result is as follows. We observe numerically that during the quench, the two-body parameters change by $< 1\%$ of their ground-state values while the one-body field undergoes order-unity changes. This motivates freezing u_2 at its ground-state value.

When the variational degree of freedom is the continuous field $u_1(\mathbf{r}, t)$, the TDVP logarithmic derivative becomes the microscopic density operator:

$$O(\mathbf{r}; \mathbf{X}) = \frac{\delta \ln \Psi}{\delta u_1(\mathbf{r})} = \sum_i \delta^{(3)}(\mathbf{r} - \mathbf{r}_i) = \hat{\rho}(\mathbf{r}). \qquad (8)$$

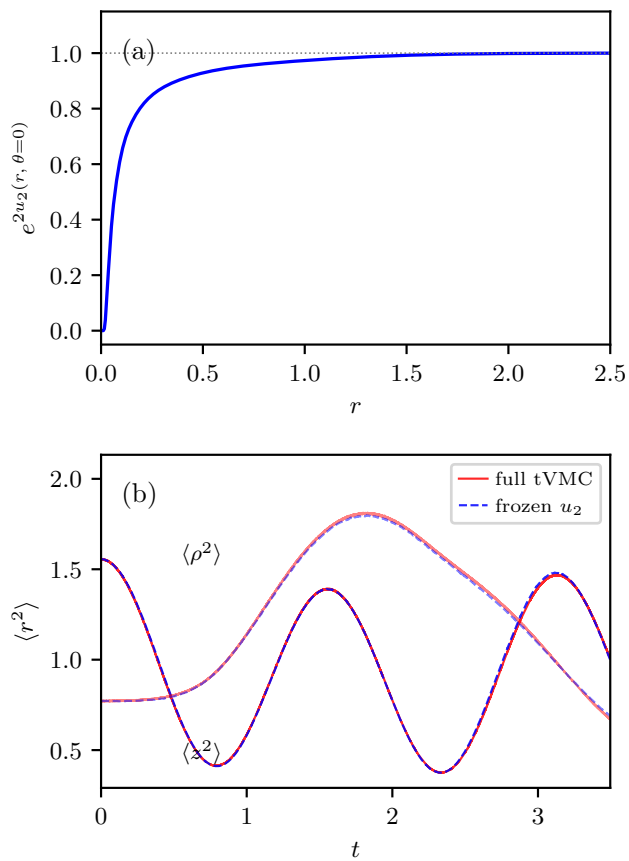

FIG. 1. (a) Leading-order HNC pair correlation $e^{2u_2(r,\theta=0)}$ for the N=8 ground state, showing a correlation hole at short range due to the shielding repulsion. (b) Breathing dynamics after the quench $\omega_\rho : 1 \to 2$ for N=128: $\langle\rho^2\rangle(t)$ and $\langle z^2\rangle(t)$ from full tVMC (red) and frozen-u_2 tVMC (blue dashed), validating the frozen-u_2 approximation.

Because u_2 is real and frozen, the many-body phase is purely one-body and irrotational: $\mathbf{v} = \nabla(\mathrm{Im}\, u_1)$. No backflow terms arise.

Decomposing $u_1 = a + i\theta$ (real amplitude and phase), the TDVP action reduces to the standard hydrodynamic form [12], yielding the continuity and Euler equations:

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0, \qquad (9)$$
$$\partial_t \mathbf{v} + \nabla\Big(\tfrac{1}{2} v^2 + V_{\rm tr} + \mu_{u_2}[\rho]\Big) = 0, \qquad (10)$$

where $\mu_{u_2}[\rho] = \delta E_{\rm int}[\rho; u_2]/\delta\rho$ is the effective chemical potential determined by the frozen correlations.

In the local-density approximation (LDA), the internal energy takes the form $E_{\rm int} = \int \rho\, \varepsilon(\rho)\, d^3 r$, where $\varepsilon(\rho)$ is the energy per particle of the *uniform* system at density ρ with the same frozen u_2. The chemical potential becomes a local function:

$$\mu(\rho) = \frac{d[\rho\varepsilon(\rho)]}{d\rho} = \varepsilon + \rho\varepsilon'. \qquad (11)$$

This is structurally identical to GP theory, except that the usual contact coupling gn is replaced by $\mu(\rho)$, which encodes the full effect of the realistic, anisotropic interaction through the frozen correlations.

The near-stationarity of u_2 during the quench is visualized directly in Fig. 2, which shows the change in the leading-order HNC pair correlation Δe^{2u_2} at four time snapshots. The variation remains below ~ 0.03 throughout (compared to peak values ~ 1), confirming that the pair correlations are effectively frozen.

Equation of state and emergent many-body interactions.—We extract $\varepsilon(\rho)$ from a one-time VMC calculation on the uniform system (periodic boundary conditions) with u_2 frozen at its trapped ground-state values. Scanning 20–30 density values logarithmically from $\rho = 0.05$ to $\rho_{\max}$ (3.0 for $N = 8$; 15.0 for $N = 128$) with 8192 walkers, we fit the results to

$$\varepsilon(\rho) = a_1\rho + a_2\rho^2 + \cdots , \tag{12}$$

yielding the chemical potential $\mu(\rho) = 2a_1\rho + 3a_2\rho^2 + \cdots$.

The physical content of this expansion is revealing. Comparing with the standard GPE nonlinearity $\mu = g\rho$, the leading term $2a_1\rho$ corresponds to a *renormalized two-body coupling* $g_{\rm eff} = 2a_1$, which absorbs the effect of the frozen pair correlations into an effective contact interaction. The next-order term $3a_2\rho^2$, however, has the structure of an *effective three-body interaction* with coupling strength $g_3 = 3a_2$. This is not a genuine microscopic three-body force; rather, it emerges from integrating out the density-dependent two-body correlations encoded in u_2. In the language of density-functional theory, it is the leading beyond-mean-field correction to the interaction energy [17].

For the shielded dipolar interaction studied here, we find $g_3/g_{\rm eff}^2 \sim 0.1$–$0.2$, indicating that the three-body term provides a 10–20% correction at the equilibrium density. For $N = 8$, a cubic fit (a_1=0.102, a_2=0.023, a_3=−0.007) is preferred by AIC, adding a further (small, negative) four-body correction; for $N = 128$, a quadratic (a_1=0.111, a_2=0.001) suffices and is unconditionally stable ($d\mu/d\rho > 0$ for all ρ). The EOS is nearly linear at high density ($a_2 \ll a_1$), reflecting the long-range character of the shielded dipolar interaction.

The VMC EOS for two interaction strengths is shown in Fig. 3. The leading coefficient a_1 scales linearly with C_3, while a_2 captures the density-dependent beyond-mean-field corrections.

The generalized GPE (14) with $\mu(\rho) = 2a_1\rho + 3a_2\rho^2$ can therefore be understood as a *two-plus-three-body* GPE:

$$i\partial_t\psi = \Big[-\tfrac{1}{2}\nabla^2 + V_{\rm tr} + g_{\rm eff}|\psi|^2 + g_3|\psi|^4 \Big]\psi, \tag{13}$$

where both couplings are microscopically determined from VMC rather than fitted to experiment. This provides a concrete, first-principles realization of the extended GPE with higher-order nonlinearities that has been discussed phenomenologically in the context of quantum droplets [18] and dipolar condensates [19, 20].

GPE solver.—We solve the generalized GPE

$$i\partial_t\psi = \Big[-\tfrac{1}{2}\nabla^2 + V_{\rm tr} + \mu(|\psi|^2) \Big]\psi, \tag{14}$$

where $\psi = \sqrt{\rho}\, e^{i\theta}$ and the Laplacian provides both quantum pressure and flow kinetic energy. The equation is discretized on a cylindrical (r, z) grid exploiting azimuthal symmetry, with a fourth-order finite-difference Laplacian (including L'Hôpital's rule at $r = 0$) and fourth-order Runge–Kutta time integration under a dispersive CFL condition $\Delta t \le C\Delta x^2/\pi$ ($C = 0.3$) [12].

The initial state is obtained by imaginary-time propagation to self-consistency: $\psi \to \psi - d\tau\, \hat{H}\psi$, normalized to N particles at each step, until the chemical potential converges to $\delta\mu < 10^{-8}$.

Results for $N = 8$.—The GPE ground state reproduces the frozen-u_2 tVMC moments to 0.7% and the energy to 0.15% (Table S1 [12]). The breathing frequency $\omega_{\rm br} = 4.19 \approx 2\omega_\rho$ is identical between tVMC and GPE, and the oscillation amplitudes of $\langle\rho^2\rangle$ and $\langle z^2\rangle$ agree to within 1%. GPE conservation is maintained at the 10^{-3} level; the tVMC energy (measured post-quench) fluctuates by $\sim 5\%$ due to stochastic noise but shows negligible drift ($< 0.01\%$).

The quantum pressure term $-\frac{1}{2}\nabla^2\psi/\psi$ is essential: without it, the dynamics is qualitatively incorrect, as the interaction energy comprises only $\sim 3.5\%$ of the total. The fourth-order Laplacian is also critical, improving the frequency from 3.99 (second-order) to 4.19 (matching tVMC).

Results for $N = 128$.—To test the framework beyond the weakly interacting regime, we scale to $N = 128$ where the interaction energy constitutes $\sim 21\%$ of the total. The EOS is extended to $\rho_{\max} = 15$ to accommodate the higher peak densities ($\rho_{\rm peak} \approx 29$) reached during compression.

The GPE ground state now matches the tVMC moments to 0.35% ($\langle\rho^2\rangle$) and 1.05% ($\langle z^2\rangle$), with a 2.8% energy difference attributable to the LDA approximation at the higher interaction strength. Remarkably, the breathing frequency $\omega_{\rm br} = 4.19$ is again identical between tVMC and GPE [Fig. 4(a)], and the $\langle\rho^2\rangle(t)$ oscillation amplitudes agree to $\sim 2\%$ (Table S1 [12]).

The GPE simulation requires ~ 70 minutes on a single CPU core (including imaginary-time ground-state convergence), compared to ~ 6 hours on a modern GPU for the full tVMC—a reduction of more than an order of magnitude in wall time, and several orders in total floating-point operations.

Discussion.—The success of the frozen-u_2 hydrodynamic reduction rests on a physical separation of scales: short-range pair correlations, governed by the interaction potential on lengths $r \lesssim a_{\rm ho}$, equilibrate much faster than the global density redistribution driven by the trap

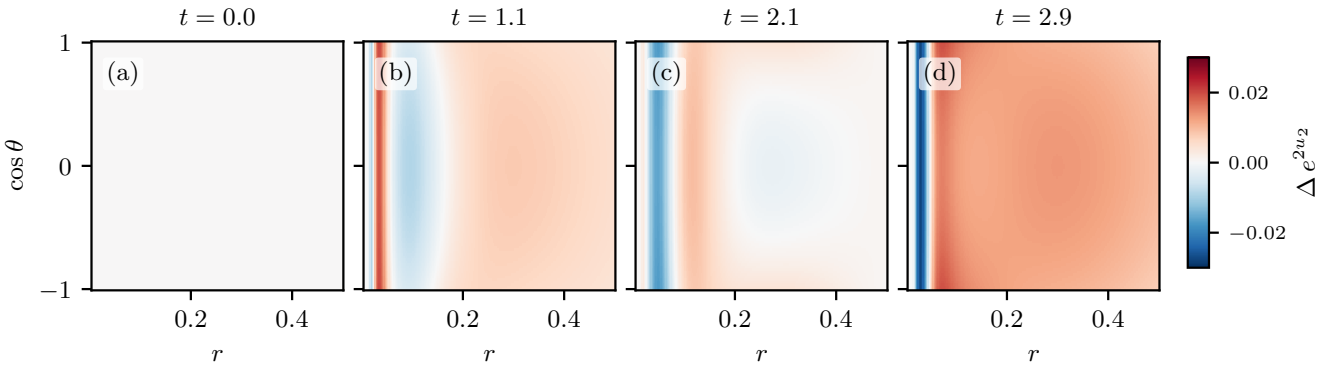

FIG. 2. Change in the leading-order HNC pair correlation, $\Delta e^{2u_2} = e^{2u_2(t)} - e^{2u_2(0)}$, during breathing dynamics for N=128. The pair correlations remain nearly frozen ($|\Delta e^{2u_2}| \lesssim 0.03$) while the one-body density undergoes order-unity changes.

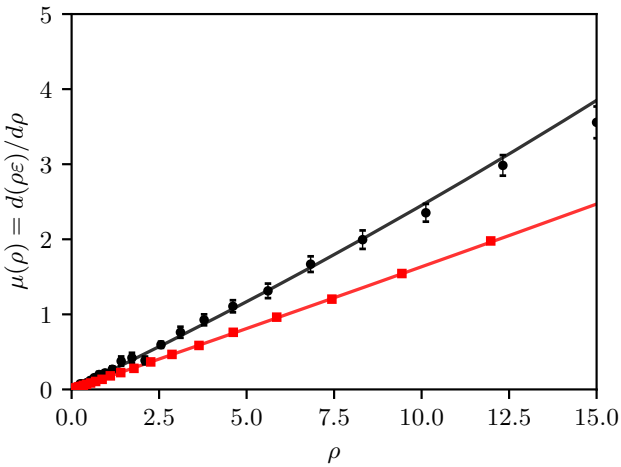

FIG. 3. Chemical potential $\mu(\rho) = d(\rho\varepsilon)/d\rho$ from uniform-system VMC for C_3=10^{-3} (circles, black fit) and C_3=10^{-4} (squares, red fit). Data points are obtained by numerical differentiation of the VMC energy; lines show polynomial fits used in the GPE.

quench. This is manifested in the tVMC as $< 1\%$ variation of the u_2 parameters during the quench, while u_1 changes by order unity.

The framework has two key advantages over standard GP theory. First, the EOS is *microscopically derived*: it incorporates the full anisotropic, density-dependent correlations of the shielded interaction through the frozen Jastrow factor, without relying on a scattering-length approximation. Second, the derivation from the TDVP is *systematic*: the frozen-u_2 approximation is a controlled limit of the full many-body variational dynamics, not an *ad hoc* mean-field assumption.

The emergence of an effective three-body coupling g_3 from the frozen pair correlations [Eq. (13)] deserves particular emphasis. In recent work on dipolar quantum droplets [18, 19], the beyond-mean-field Lee–Huang–Yang (LHY) correction is typically added *ad hoc* as a $|\psi|^4$ nonlinearity with a coupling estimated from perturbation theory. Our framework provides a non-perturbative, microscopically grounded alternative: the g_3 coupling is extracted directly from VMC and automatically includes the full anisotropy and density dependence of the shielded interaction. For the parameters studied here, $g_3 > 0$ (repulsive), consistent with the stabilizing role of beyond-mean-field corrections in dipolar systems. Extending the EOS extraction to different interaction strengths and geometries would map out how g_3 evolves across the phase diagram—information that is difficult to obtain from perturbative approaches alone.

As a demonstration of predictive power, we apply the framework to $N = 10{,}000$ particles with a weaker interaction ($C_3 = 10^{-4}$), a regime entirely inaccessible to direct tVMC. The GPE predicts sustained breathing oscillations over $t = 10$ trap periods [Fig. 4(b)], providing quantitative predictions for upcoming molecular BEC experiments.

The $\sim 3\%$ energy offset at $N = 128$ suggests that gradient corrections to the LDA [21] could further improve accuracy. The framework naturally extends to other collective modes (scissors, quadrupole), different quench protocols, and mixed-dimensional geometries relevant to current molecular experiments [1, 22].

Conclusion.—We have demonstrated that correlated many-body dynamics from tVMC maps *exactly* onto superfluid hydrodynamics when two-body correlations are frozen, with the equation of state determined by a single VMC calculation. The resulting generalized GPE reproduces the breathing dynamics of microwave-shielded dipolar bosons from $N = 8$ to $N = 128$ with $< 1\%$ frequency accuracy and $\sim 2\%$ amplitude accuracy, at drastically reduced computational cost. This provides a practical, quantitatively reliable tool for interpreting and predicting collective-mode experiments in the rapidly expanding field of shielded molecular quantum gases.

Computations were performed using JAX [16] for

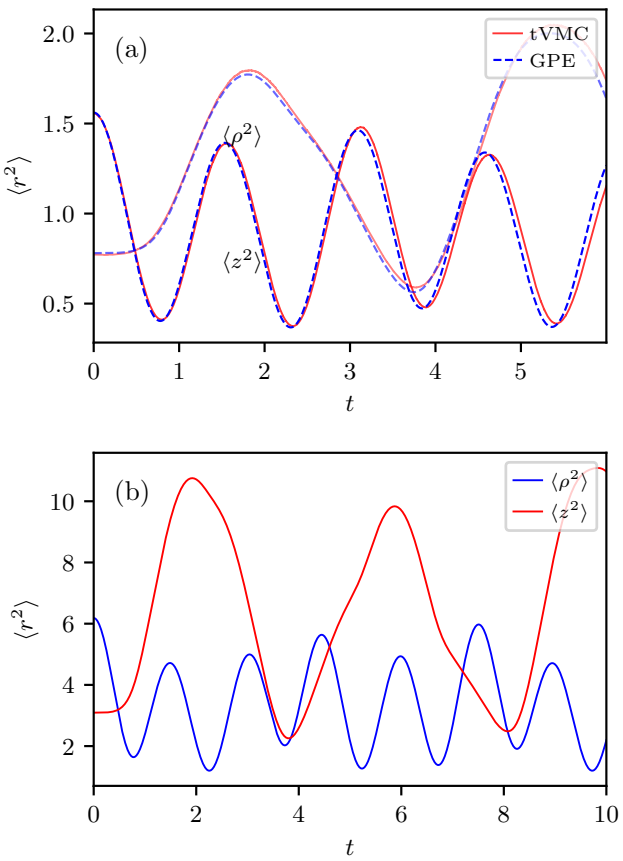

FIG. 4. GPE dynamics from the VMC-derived EOS. (a) N=128 validation: $\langle\rho^2\rangle(t)$ and $\langle z^2\rangle(t)$ from frozen-u_2 tVMC (red) and GPE (blue), showing $< 1\%$ frequency agreement ($\omega_{\rm br}$=4.19). (b) N=10,000 prediction with the weak-interaction EOS (C_3=10^{-4}), extrapolating to experimentally relevant system sizes inaccessible to direct tVMC.

GPU-accelerated VMC and NumPy/SciPy for the hydrodynamic solver.

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